

# SECTION 4.3 SUPPLEMENTAL EXERCISES

## INTRODUCTION TO COMPLEX ANALYSIS

(MATH 4230-40), SPRING 2022

The following are indicative of the level of difficulty of problems from Section 4.3 that you will find on Exam II (there will not be more than one or two problems from this section).

### SECTION 4.3 - EVALUATION OF DEFINITE INTEGRALS

1. Evaluate  $\int_{-\infty}^{\infty} \frac{x^2 - 3i}{x^4 + 5} dx$ .

Note that  $f(z) = \frac{z^2 - 3i}{z^4 + 5}$  is a rational function with a denominator having no real zeros and of degree at least two greater than that of the numerator. As such, Proposition 4.3.6 yields that

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum \{\text{residues of } f \text{ in } \mathcal{H}\} = 2\pi i \sum \{\text{residues of } f \text{ in } \mathcal{L}\},$$

where  $\mathcal{H}$  and  $\mathcal{L}$  are the upper and lower half planes, respectively. The  $4^{th}$  roots of  $-5$  are given by

$$z_1 = \frac{-(1+i)\sqrt[4]{5}}{\sqrt{2}}, \quad z_2 = \frac{(1-i)\sqrt[4]{5}}{\sqrt{2}}, \quad z_3 = \frac{-(1-i)\sqrt[4]{5}}{\sqrt{2}}, \quad z_4 = \frac{(1+i)\sqrt[4]{5}}{\sqrt{2}}.$$

As each is a simple pole of  $f$ , we may compute the residues as

$$\text{Res}(f; z_k) = \left. \frac{z^2 - 3i}{4z^3} \right|_{z=z_k} = \frac{1}{4z_k} - \frac{3i}{4z_k^3}.$$

Noting that  $z_1, z_2 \in \mathcal{L}$ , we have

$$\int_{-\infty}^{\infty} \frac{x^2 - 3i}{x^4 + 5} dx = 2\pi i \left( \frac{z_1^2 - 3i}{4z_1^3} + \frac{z_2^2 - 3i}{4z_2^3} \right) = \frac{(\sqrt{5} - 3i)\pi}{5^{3/4}\sqrt{2}}.$$

Just as well, we could also note that  $z_3, z_4 \in \mathcal{H}$  and produce

$$\int_{-\infty}^{\infty} \frac{x^2 - 3i}{x^4 + 5} dx = 2\pi i \left( \frac{z_3^2 - 3i}{4z_3^3} + \frac{z_4^2 - 3i}{4z_4^3} \right) = \frac{(\sqrt{5} - 3i)\pi}{5^{3/4}\sqrt{2}}.$$

2. Evaluate  $\int_{-\infty}^{\infty} \frac{\cos(4x) - i \sin(4x)}{6x^2 + 2} dx$ .

To begin, observe that:

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{\cos(4x) - i \sin(4x)}{6x^2 + 2} dx &= \int_{-\infty}^{\infty} \frac{\cos(4x)}{6x^2 + 2} dx - i \int_{-\infty}^{\infty} \frac{\sin(4x)}{6x^2 + 2} dx \\ &= \int_{-\infty}^{\infty} \operatorname{Re} \left( \frac{e^{4ix}}{6x^2 + 2} \right) dx + i \int_{-\infty}^{\infty} \operatorname{Im} \left( \frac{e^{4ix}}{6x^2 + 2} \right) dx \\ &= \operatorname{Re} \left( \int_{-\infty}^{\infty} \frac{e^{4ix}}{6x^2 + 2} dx \right) + i \operatorname{Im} \left( \int_{-\infty}^{\infty} \frac{e^{4ix}}{6x^2 + 2} dx \right) \\ &= \int_{-\infty}^{\infty} \frac{e^{4ix}}{6x^2 + 2} dx \end{aligned}$$

Writing  $f(x) = \frac{p(x)}{q(x)} = \frac{1}{6x^2 + 2}$ , note that  $q$  is a polynomial of degree greater than  $p$  and that  $q$  has no zeros on the real axis. As such, this is an integral of the form  $\int_{-\infty}^{\infty} f(x)e^{-i\omega x} dx$  satisfying part (iii) of Proposition 4.3.9, and we have

$$\int_{-\infty}^{\infty} f(x)e^{i\omega x} dx = 2\pi i \sum \left\{ \text{residues of } f(z)e^{-i\omega z} \text{ in } \mathcal{H} \right\}.$$

So, as  $f(z)e^{4iz} = \frac{e^{4iz}}{6x^2 + 2}$  has simple poles at  $z = \pm \frac{i}{\sqrt{3}}$ , we get that

$$\operatorname{Res} \left( \frac{e^{4iz}}{6x^2 + 2}; \frac{i}{\sqrt{3}} \right) = \frac{e^{-4/\sqrt{3}}}{\frac{12i}{\sqrt{3}}} = \frac{ie^{-4/\sqrt{3}}}{4\sqrt{3}}.$$

Therefore, we conclude that

$$\int_{-\infty}^{\infty} \frac{\cos(4x) - i \sin(4x)}{6x^2 + 2} dx = 2\pi i \operatorname{Res} \left( \frac{e^{4iz}}{6x^2 + 2}; \frac{i}{\sqrt{3}} \right) = \frac{\pi e^{-4/\sqrt{3}}}{2\sqrt{3}}.$$

3. Compute the Cauchy principal value of  $\int_{-\infty}^{\infty} \frac{z^2 - 3i}{z^5 + 5} dz$

Write  $f(z) = \frac{p(z)}{q(z)}$  with  $p(z) = z^2 - 3i$  and  $q(z) = z^5 + 5$ . Observe that  $p$  and  $q$  are polynomials and the degree of  $q$  is at least two greater than the degree of  $p$ . However,  $q$  does indeed have a zero of order one on the real axis (i.e.  $f$  has a simple pole on the real axis), so Proposition 4.3.6 does not apply. Instead, letting  $\mathcal{H}_0$  denote the open upper half plane (i.e.  $\mathcal{H}_0 = \{z \in \mathbb{C} : \text{Im}z > 0\}$ ) we make use of Proposition 4.3.11 to see that

$$\text{P.V.} \int_{-\infty}^{\infty} f(z) dz = 2\pi i \sum \{\text{residues of } f \text{ in } \mathcal{H}_0\} + \pi i \sum \{\text{residues of } f \text{ on the real axis}\}.$$

The zeros of  $q(x)$  are the 5<sup>th</sup> roots of  $-5$ , which are given by

$$z_k = -\sqrt[5]{5} \left( \cos\left(\frac{2k\pi}{5}\right) + i \sin\left(\frac{2k\pi}{5}\right) \right), \quad k = 0, 1, 2, 3, 4.$$

Of these,  $z_0$  lies on the real axis, while  $z_3$  and  $z_4$  lie in  $\mathcal{H}_0$ . As all are simple poles of  $f$ , we have

$$\text{Res}\left(\frac{z^2 - 3i}{z^5 + 5}; z_k\right) = \frac{z_k^2 - 3i}{5z_k^4}.$$

Using this and computing, we get

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{z^2 - 3i}{z^5 + 5} dz = 2\pi i \left( \text{Res}\left(\frac{z^2 - 3i}{z^5 + 5}; z_3\right) + \text{Res}\left(\frac{z^2 - 3i}{z^5 + 5}; z_4\right) \right) + \pi i \text{Res}\left(\frac{z^2 - 3i}{z^5 + 5}; z_0\right) = -\sqrt[5]{5}\pi \sqrt{5 + 2\sqrt{5}}$$